

Rustom Murgai Dubash

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Education

University of Pennsylvania, MSE Computer and Information Science (*Submatriculation*)

- Graduate-level coursework completed concurrently with undergraduate study.

Haverford College, BSc Computer Science and Economics

Graduation: May 2027

- GPA: 3.93/4.0 | Varsity Cricket Team

Publications

Rustom M. Dubash and Sorelle A. Friedler. 2026. **Self-Promotion in LLM Recommendations**. In *18th ACM Web Science Conference (WebSci '26)*, May 26–29, 2026, Braunschweig, Germany. ACM.

Experience

AI Solutions Engineer Intern, ContinuumServe, Marietta, GA

June 2026 – Present

- Building a **Financial Intelligence Agent** using **OpenClaw**, using a 4-agent pipeline: a **Financial Data Collection Agent** ingesting ERP/CSV/Excel sources, a **Financial Analysis Agent** detecting anomalies and variance, an **Executive Reporting Agent** generating KPI dashboards, and an **Accounting Knowledge Assistant** enabling semantic retrieval.
- Integrated **vector databases**, **OpenAI/Claude APIs**, and **prompt engineering** to automate P&L reporting, balance sheet analysis, and executive summary generation across ERP and finance systems.

Teaching Assistant: Introduction to Computer Science & Data Science, Haverford College

Sept 2025 – Jan 2026

- Held weekly office hours to support **80+ students'** learning in Python programming, **algorithms**, and **data analysis**.

Applied Data Institute Scholar '25, Equitech Futures

Sept 2025 – Dec 2025

- Selected as one of **16 fellows** for a **10-week**, fully funded (**\$9,500**) global fellowship in leadership, data science, and AI.
- Developed improved metrics based on water utility data - Key Performance Indicators (KPIs)- for water utility managers in Uganda, Cameroon, Lesotho, and Malawi, impacting **5M+ citizens** in partnership with **Athena Infonomics**.
- Developed dashboards covering **20+ Key Performance Indicators** and improved data accessibility by integrating an **AI-driven RAG system** using **ChromaDB**, **text embeddings**, and an **API-wrapped chatbot** interface.

Machine Learning Researcher, Mathieson Lab, Haverford College

May 2025 - Aug 2025

- Designed, trained, and evaluated **Transformer sequence models** in PyTorch, conducting architecture-level analysis of **attention mechanisms**, context length, and inductive biases on **next-token prediction tasks**.
- Implemented long-context alternatives to **quadratic self-attention**, including hybrid convolutional and **sparse-attention architectures** such as HyenaDNA, increasing context length by $12\times$ baseline Transformer without compromising accuracy.
- Built scalable training and preprocessing pipelines for a 21 GB genomic DNA sequence corpus, using memory-efficient **data loading**, **multiprocessing**, and **GPU acceleration** to support large-scale sequence modeling.

Research Assistant, Dept. of Economics, National University of Singapore

June 2024 - Aug 2024

- Led data compilation for a study on salary incentives and political corruption in Indian State Legislatures.
- Built and deployed web-scraping pipelines to extract **10K+ financial and news records** for corruption analysis.

Selected Projects and Research

Residual-Boosted Transformer Ensembles (**Compute-Accuracy Tradeoffs**), TensorFlow/Keras, Ensembling

- Implemented **residual boosting** over **ensemble logits** by fitting **probability residuals** with weak Transformers, achieving **93% accuracy** on Yelp Polarity benchmark dataset with a 10M parameter budget (SOTA: 98% accuracy, 355M parameters).

Morphology-Focused Learning for Histopathology, TensorFlow, Grad-CAM, PCA/KMeans, Robustness testing

- Diagnosed spurious correlations in a histopathology CNN by computing **Gradient-weighted Class Activation Mapping** heatmaps, revealing when predictions are driven by stain intensity and tissue boundaries rather than morphology.
- Trained a **morphology-focused model** using color jitter and grayscale augmentation, achieving identical performance on RGB vs. grayscale images (**acc. 0.84**), indicating reduced reliance on color and stronger dependence on structural features.

Financial Sentiment Stock Predictor, Python, XGBoost, VADER, FinBERT, yfinance

- Built a ML model using **financial news sentiment** and historical price data to predict next-day stock returns
- Engineered sentiment scores using FinBERT and financial lexicons across **50k+ headlines** from major outlets
- Automated daily data ingestion with the yfinance API, enabling **end-to-end retraining** and **backtesting**

Skills

Programming: Python, Java, JavaScript, HTML, CSS, ReactJS, NodeJS, SQL

Frameworks/Libraries: PyTorch, TensorFlow, pandas, numpy, scikit-learn, Flask, Streamlit, Pygame, Biopython, ChromaDB

Tools/Platforms: Git, VSCode, Render, LaTeX, Excel, Microsoft 365, STATA, BLAST, Phobius, Interpro, UniProt